FYS5419/9419 March 25

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DFT : N^2 complex multiplications

$N(N-1)$ complex additions

FLOPS $\sim O(N^2)$

FFT : $\frac{1}{2} N \log_2 N$ complex multiplications

$N \log_2 N$ complex additions

FLOPS $\sim O(N \log_2 N)$

QFT : Flops $\sim O(n^2)$

n = # qubits

$N = 2^n$

$(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$

$n = 100$

DFT : $O(2^{200}) \sim 10^{60}$

FFT : $O(N \log_2 N)$

$n 2^n \log_2 2 \sim n 2^n$

QFT : $O(n^2) \sim 10^4 \sim 100 \times 10^{30}$

$$N = 2^n \quad (n=2,\ |00\rangle, |01\rangle, |10\rangle, |11\rangle)$$

$$QFT_N |x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega_N^{xk} |k\rangle$$

$$\omega_N = e^{2\pi i/N}$$

$$x = x_{n-1} 2^{n-1} + x_{n-2} 2^{n-2} + \cdots + x_1 \cdot 2^1 + x_0 \cdot 2^0$$

$$|x\rangle = |x_{n-1} \ldots x_0\rangle$$

$$(|x_0 x_1 \ldots x_{n-1}\rangle)$$

$$x_j = \{0, 1\}$$

$x_1 \quad x_0 \qquad x_i = \{0, 1\}$

Binary Float

$$0.x_1 x_0 = \frac{x_1}{2} + \frac{x_0}{4}$$

most significant bit

least significant bit

$$0.x_m x_{m-1} \ldots x_0 =$$

$$\frac{x_m}{2} + \frac{x_{m-1}}{2^2} + \ldots \frac{x_0}{2^{m+1}}$$

$$|x\rangle = |x_1 x_0\rangle \quad (x = 2x_1 + x_0)$$

integer

$$QFT_{N=4}\,|x\rangle = \frac{1}{2}\sum_{k=0}^{3} e^{2\pi i x k/4}\,|k\rangle$$

$$k = 2k_1 + k_0 \qquad k_1, k_0 \in \{0,1\}$$

$$QFT_4\,|x_1 x_0\rangle = \frac{1}{2}\sum_{k_1 k_0 \in \{0,1\}} e^{2\pi i (2x_1 + x_0)(2k_1 + k_0)/4}\,|k_1 k_0\rangle$$

$$e^{2\pi i x_1 k_1} = 1 \quad \text{for integer}$$

$$x_1 \cdot k_1$$

$$e^{2\pi i x k/4} = e^{2\pi i \left(\frac{x_1 k_0}{2} + \frac{x_0 k_1}{2} + \frac{x_0 k_0}{4}\right)}$$

$$e^{2\pi i x k/4} = e^{2\pi i k_1 \left(\frac{x_0}{2}\right)} \times e^{2\pi i k_0 \left(\frac{x_1}{2} + \frac{x_0}{4}\right)}$$

$$QFT_4 \, |x_1 x_0\rangle = \frac{1}{2} \sum_{k_1 k_0} e^{2\pi i k_1 (0.x_0)} \times$$

$$e^{2\pi i k_0 (0.x_1 x_0)} |k_1 k_0\rangle$$

$$= \frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i\, 0.x_0} |1\rangle\right) \otimes$$

$$\frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i\, 0.x_1 x_0} |1\rangle\right)$$

Key point

- First factor depends only on x_0
- Second depends on $x_1 x_0$

Gates

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$$

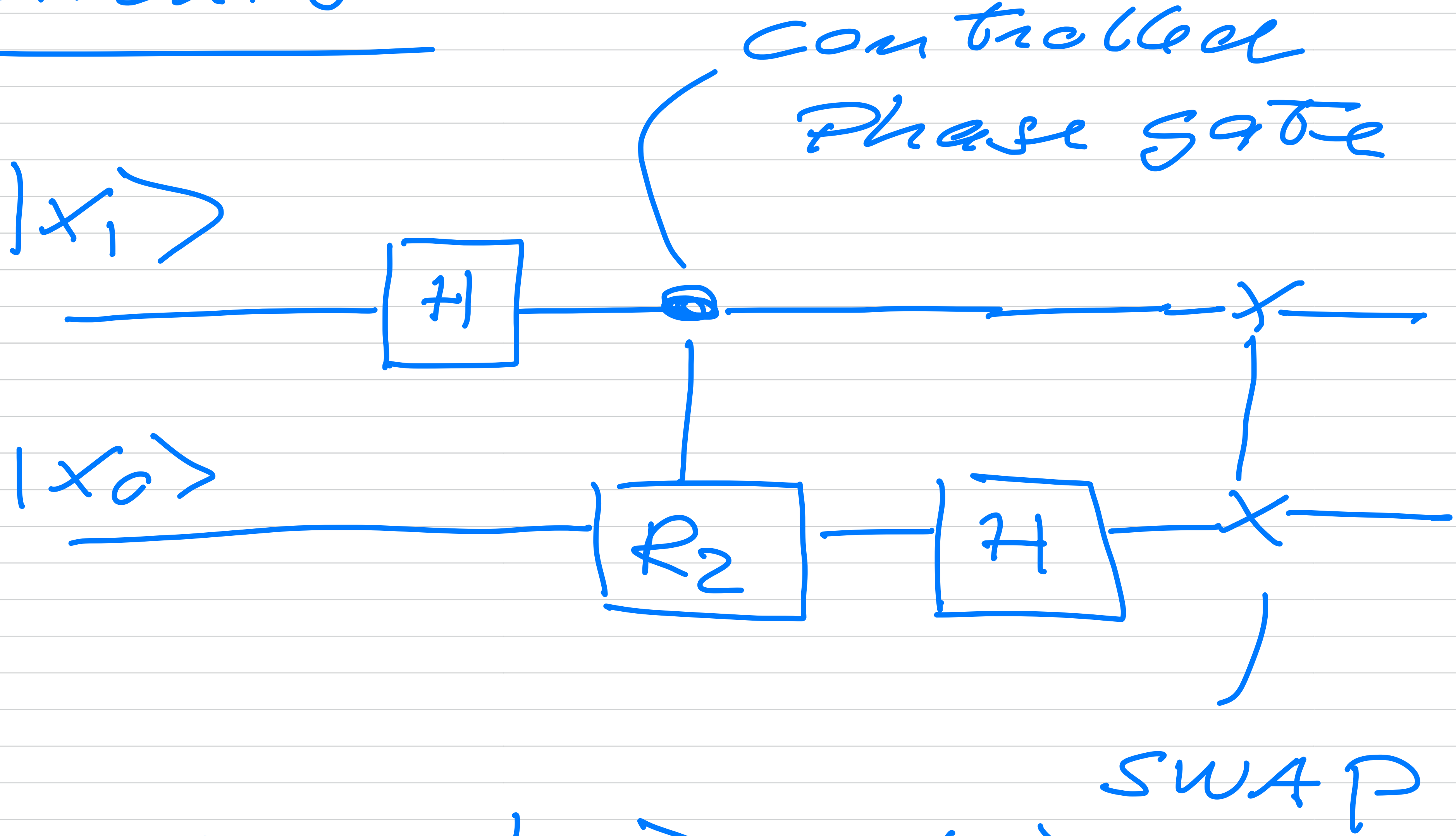
(Hadamard

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Phase gate

$$R_2 = \begin{bmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & e^{\pi i/2} \end{bmatrix}$$

Circuit



$\text{SWAP}|01\rangle = |10\rangle$

$\quad$ — " — $|10\rangle = |01\rangle$

$\quad$ — " — $|11\rangle = |11\rangle$ same with $|00\rangle$

Step 1 : apply H to first qubit

start with $|x_1 x_0\rangle$

$$|x_1 x_0\rangle \to \frac{1}{\sqrt{2}}\left(|0\rangle + e^{i\pi x_1}|1\rangle\right) \otimes |x_0\rangle$$

$$e^{i\pi x_1} = (-1)^{x_1}$$

$$\frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i x_1/2}|1\rangle\right) \otimes |x_0\rangle$$

Step 2 Controlled Phase R_2 adds a phase $e^{i\pi/2}$ only when both qubits are in $|1\rangle$

$|1\rangle \otimes |x_0\rangle$ picks up an extra factor $e^{2\pi i x_0/4}$

$$\frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i\left(\frac{x_1}{2}+\frac{x_0}{4}\right)}|1\rangle\right) \otimes |x_0\rangle$$

$$= \frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i\, 0.x_1x_0}|1\rangle\right) \otimes |x_0\rangle$$

Apply H to the second qubit

$$|x_0\rangle \rightarrow \frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i x_0/2}|1\rangle\right)$$

Hence the full state is:

$$\underbrace{\frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i\, 0.x_1x_0}|1\rangle\right)}_{H + R_2 \text{ on first}} \otimes \underbrace{\frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i\, 0.x_0}|1\rangle\right)}_{H \text{ on 2nd}}$$

qubits are in reversed order
need a swap gate to
bring back from

$$|k_1 k_0\rangle \longleftrightarrow |k_0 k_1\rangle$$

The phases in 2-qubit case
are

$$0.x_0 = x_0/2$$

$$0.x_1x_0 = x_1/2 + x_0/4$$

$$0.x_2x_1x_0 = \frac{x_2}{2} + x_1/4 + \frac{x_0}{8}$$

$|x_2x_1x_0\rangle$ instead of $|x_0x_1x_2\rangle$